

Enhancing the Accuracy of Predicting Kyphosis Disease Through Machine Learning Approaches XgBoost and Decision Tree

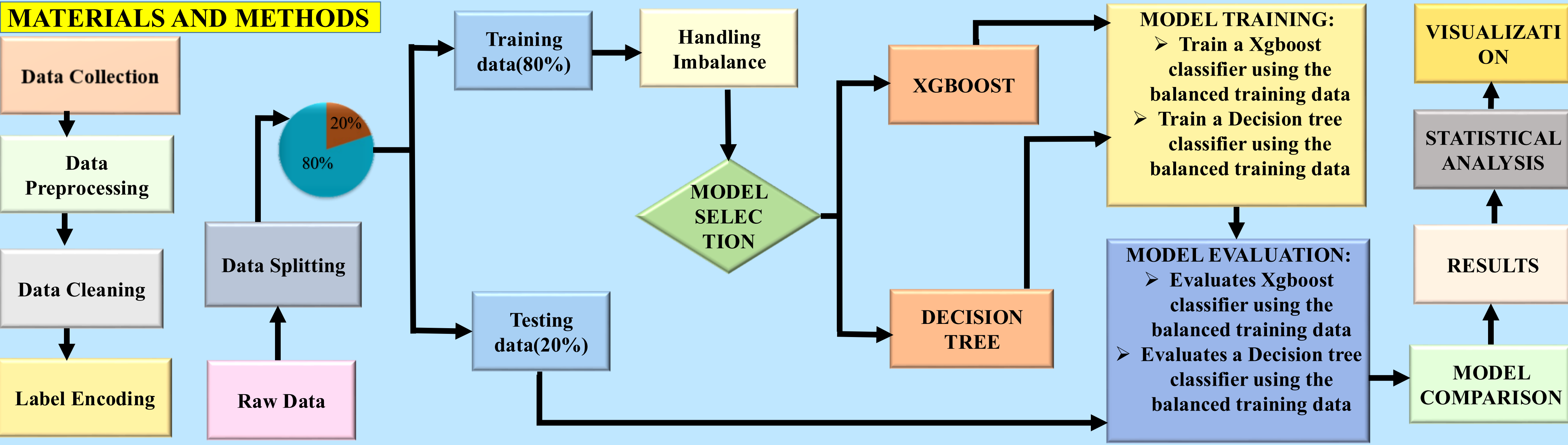
INTRODUCTION

- The aim is to leverage Xgboost and Decision tree algorithms to identify the most accurate machine learning approach for kyphosis prediction.
- Accurate prognosis plays a vital role in healthcare by facilitating proactive intervention, leading to better patient outcomes and reduced healthcare costs.
- The application of Xgboost and Decision tree in predicting kyphosis disease through machine learning enables accurate diagnosis, early intervention, and personalized treatment plans, revolutionizing healthcare in this field.
- Xgboost is a speedy algorithm that handles missing data effectively, while Decision tree algorithm is transparent and adaptable to various data types.
- The kyphosis detection dataset utilized is sourced from Kaggle, a reputable platform, Enabling seamless data sharing and supporting data driven investigation.



Fig. 1:Normal and affected spine by kyphosis.

MATERIALS AND METHODS



RESULTS

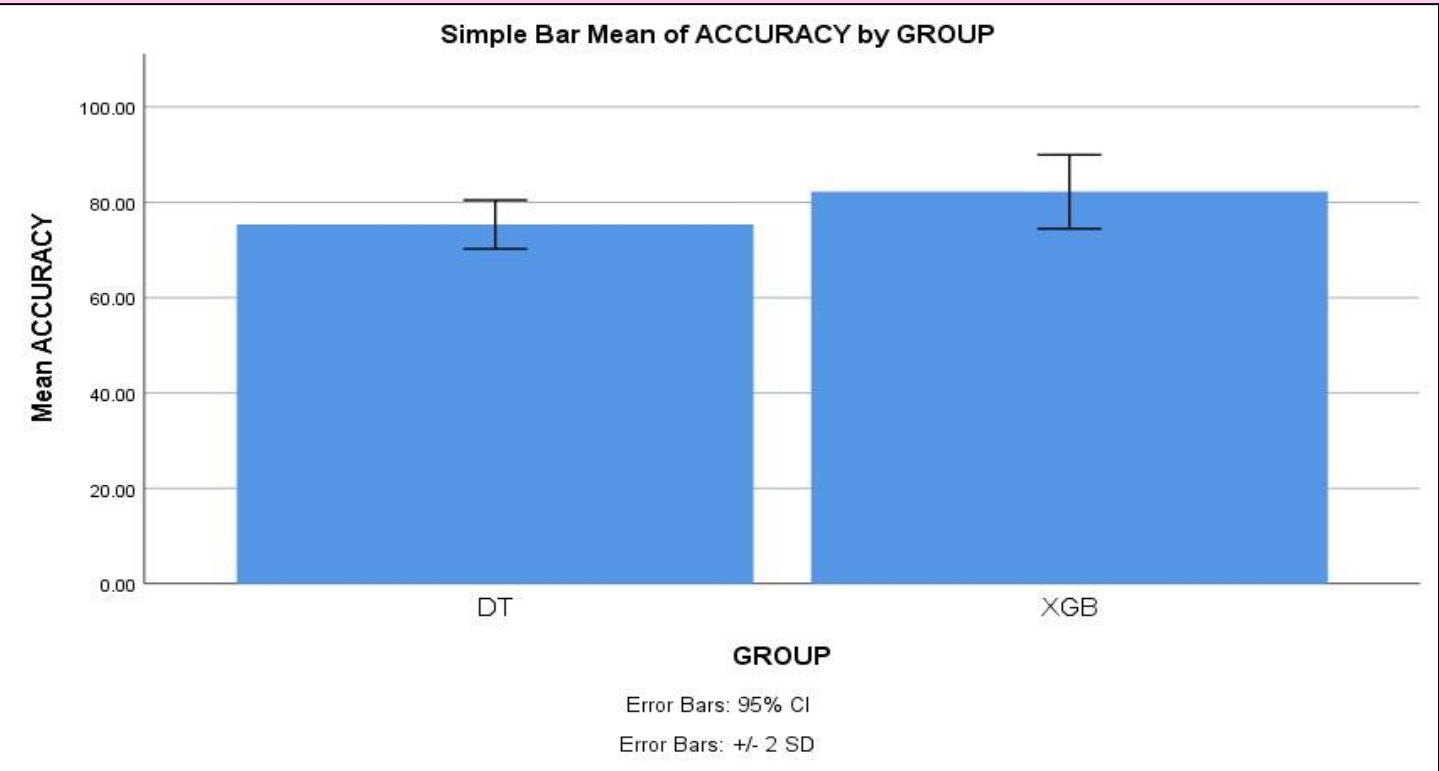


Fig. 2: Comparison of kyphosis detection between Xgboost and Decision tree.

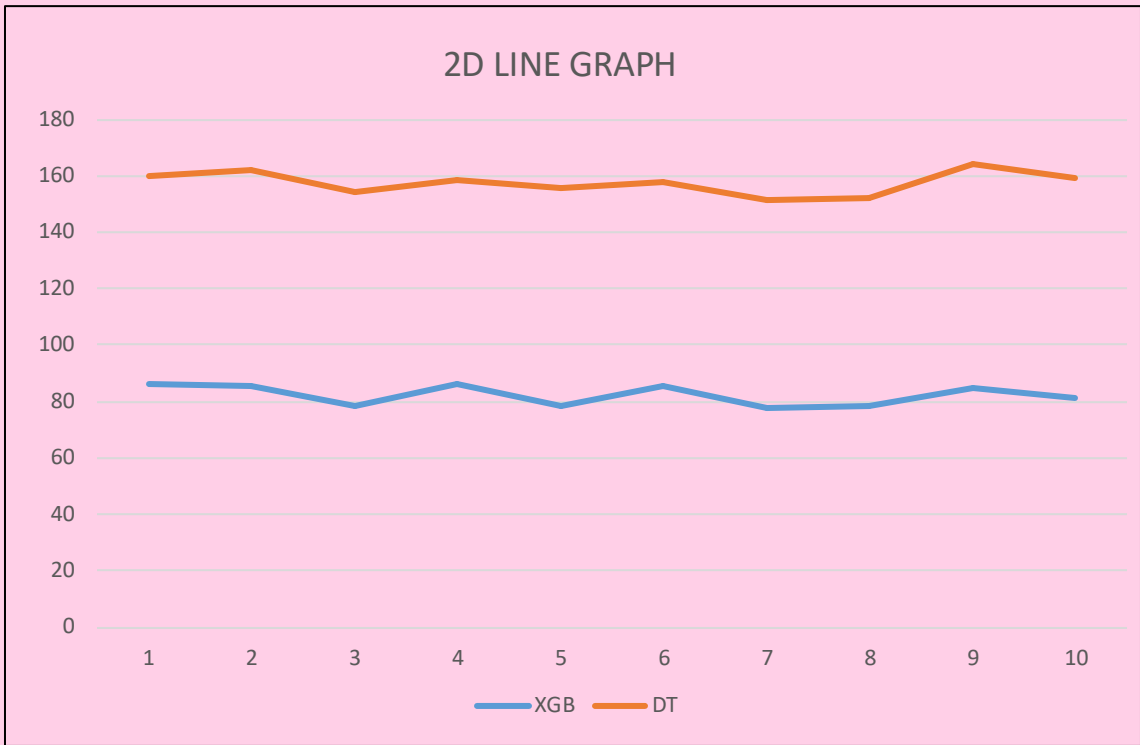


Fig. 3: Comparison of accuracy values of kyphosis detection with Xgboost and Decision tree.

GROUP		N	Mean	Std. Deviation	Std. Error Mean
ACCURACY	XGB	10	82.2140	3.87330	1.22484
	DT	10	75.3540	2.55185	0.80697

Table 1: Comparison of the mean accuracy of kyphosis detection with Xgboost and Decision tree.

DISCUSSION AND CONCLUSION

- Based on T-test statistical analysis, the significance value of $p=0.001$ (independent sample T-test $p<0.05$) is obtained and shows that there is a statistical significant difference between the group 1 and group 2 (group 1-Xgboost & group 2-Decision tree).
- The Decision Tree algorithm exhibited an accuracy rate of 76.46%, whereas the Xgboost algorithm, demonstrated an accuracy of 82.35% in predicting kyphosis disease in individuals.
- Further research can explore the integration of other machine learning algorithms and feature engineering techniques to enhance the accuracy of kyphosis prediction.
- Age, gender, and other factors can affect the accuracy of kyphosis prediction using machine learning. However, limitations arise from data availability, quality, potential bias, and the need for validation on diverse populations.
- Machine learning approaches, specifically Xgboost and Decision tree algorithms, show promise in accurately predicting kyphosis. However, more research and validation are necessary to address limitations and improve the overall effectiveness of these approaches.

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